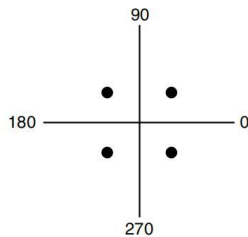


《计算机网络》作业

第三次作业：(英文版教材第二章 21, 24, 25, 26, 28, 36, 37)

21(2.5.2 Passband Transmission). A modem constellation diagram similar to Fig. 2-23 has data points at $(0, 1)$ and $(0, 2)$. Does the modem use phase modulation or amplitude modulation?



24(2.6.3 The Local Loop: Modems, ADSL, and Fiber). An ADSL system using DMT allocates $3/4$ of the available data channels to the down-stream link. It uses QAM-64 modulation on each channel. What is the capacity of the downstream link?

25(2.5.3 Frequency Division Multiplexing). Ten signals, each requiring 4000 Hz, are multiplexed onto a single channel using FDM. What is the minimum bandwidth required for the multiplexed channel? Assume that the guard bands are 400 Hz wide.

26(2.6.4 Trunks and Multiplexing). Why has the PCM sampling time been set at 125 μsec ?

28(2.1.3 The Maximum Data Rate of a Channel). Compare the maximum data rate of a noiseless 4-kHz channel using

- (a) Analog encoding (e.g., QPSK) with 2 bits per sample.
- (b) The E1 PCM system.

36(2.6.5 Switching) Compare the delay in sending an x -bit message over a k -hop path in a circuit-switched network and in a (lightly loaded) packet-switched network. The circuit setup time is s sec, the propagation delay is d sec per hop, the packet size is p bits, and the data rate is b bps. Under what conditions does the packet network have a lower delay? Also, explain the conditions under which a packet-switched network is preferable to a circuit switched network.

37(2.6.5 Switching). Suppose that x bits of user data are to be transmitted over a k -hop path in a packet switched network as a series of packets, each containing p data bits and h header bits, with $x \gg p + h$. The bit rate of the lines is b bps and the propagation delay is negligible.

What value of p minimizes the total delay?